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S-AI v6.1: a security-hardened Synthetic Executive for policy-gated multi-agent autonomy with TRL/MRL/IRL readiness evidence

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Software

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Summary

S-AI v6.1 is a security-hardened evolution of the S-AI agent swarm that adds a **Synthetic Executive**: a controller that plans *and* acts through risk-rated tool execution with explicit approval gates. The release hardens the platform for production use with SSRF protection, filesystem and shell sandboxing, bearer-token authentication, rate limiting, and a registry-bound execution engine, and it measures its own maturity with Technology Readiness Level (TRL), Manufacturing Readiness Level (MRL), and Integration Readiness Level (IRL) reporting. It retains the full S-AI capability surface — multi-agent consensus, Digital Twin persona adaptation, simulated-organoid memory, MCP Builder, Skill Creator, research mapping, multilingual support, knowledge graph, and the local-first OpenWorker harness.

Statement of need

Autonomous agent systems are typically demonstrated in demos, then fail in production because their security boundary is an afterthought: untrusted inputs can trigger SSRF, tools execute outside the sandbox, and there is no formal statement of how ready the system is for deployment. Anyone deploying an autonomous assistant needs (i) a security boundary that is designed in, not patched on, and (ii) an explicit engineering assessment language for claiming readiness. S-AI v6.1 addresses the first with layered, tested security controls and the second with TRL/MRL/IRL evidence attached to the release process itself, so a deployer can see measured readiness rather than a marketing claim.

State of the field

Existing agent orchestration research focuses on capability and coordination; guardrail frameworks gate *output*, and sandboxing layers (eQubes-style or container-based tools, e.g. OCI sandboxes and Cloudflare-style workers) gate *processes*, but few systems integrate output gating, input validation against SSRF, and a formal readiness assessment into one deployable. S-AI v6.1 positions itself in that gap: it treats execution as a security architecture (registry-bound tool registry, approval-gated risk-rating, sandboxed subprocesses, bearer-token auth, rate limiting) and couples it to a published readiness protocol. The result is a working reference for "production-grade bounded autonomy" that researchers can reproduce and extend.

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Software design

The release is organized around the security and readiness layers on top of the S-AI core. The **Synthetic Executive** turns swarm plans into a risk-sorted action list; each action is rated by the policy engine and either allowed, denied, or queued for human approval. The **hardening layer** adds SSRF-guarded HTTP fetching, a filesystem/shell sandbox that denies writes outside allowed roots, bearer-token authentication for the server component, per-route rate limits, and an execution engine that only runs tools from a registry-bound allow-list. Maturity evidence is generated by the `readiness` module, which emits TRL/MRL/IRL indicators computed from reproducible checks (tests pass, security scans pass, release artifacts build, documentation exists). Every claimed level traces to CI evidence.

Research impact statement

The distinguishing contribution of S-AI v6.1 is that *readiness claims are derived, not declared*: the TRL/MRL/IRL indicators are computed from the same automated pipelines a reviewer can re-run, connecting engineering-state reporting to open-source-software maturity evaluation. The security controls (SSRF protection, sandboxing, auth, rate limiting) are each exercised by the repository's CI workflows, making the security posture auditable. This makes the repository a useful reference for research on safe deployment of agentic systems and on automated TRL-style maturity assessment for software.

AI usage disclosure

The code, documentation, security analysis, and this paper were drafted with generative-AI assistance (code scaffolding, security review assistance, copy-editing). All assisted artifacts were reviewed, edited, and validated by the human author, who made the design decisions. No AI rendered evaluative decisions in this submission.

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References

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